Latch handle | CAD reference atlas

Photographic record of the original sample

English edition - 14 November 2026. For reconstruction of a parametric 3D model.

This atlas contains seven original views and three owner-annotated photographs. Duplicate uploads of the same views are not repeated. Each photograph is accompanied by the camera direction, visible features and intended CAD use.

Front means the broad paddle side. Rear means the side with the latch tooth, reinforcing ribs and elongated flange recess. Up, down, left and right refer to the individual photograph, not the installed mechanism.

Reference	Dimension or requirement
33.5 mm	Overall height from lug crown to lowest flange edge in the side view; photo 08.
16.5 mm	Flange height measured along its inclination; photo 09.
11.3 mm	From the flat platform with tabs to the lug crown; owner measurement.
30 mm	Paddle length from the flange step to the free end; owner measurement.
Approximately 110 degrees	Two obtuse joint angles estimated from photo 10, not measured.
Latch tooth	Almost straight, essentially parallel to the main paddle plane; no pronounced hook.

Scope and source status

Pages 1-12 translate the supplied Russian photo atlas dated 12 November 2026. Its separate cad-latch-spec.md revision 2 was referenced but was not embedded in the PDF. The English appendix consolidates the specification values and decisions preserved in the conversation; it is not a verified line-by-line translation of that unavailable Markdown file.

Photographs are perspective views and are not printed to scale. The sample tooth is damaged. Do not reproduce cracks or chips; verify the missing profile and engagement using a prototype, mating component or intact sample.

01. General view from the paddle side



Source: IMG_6558(1).JPG

View. Oblique view of the broad front face of the paddle, showing the platform and pivot lugs. The part rests on graph paper; the camera is angled to its principal planes.

Visible features. The paddle tapers towards its free end. Both rounded corners, the two drilled pivot lugs and the flange wings are visible.

CAD use. Use for the overall silhouette and relative arrangement of features. Perspective prevents direct measurement of widths and radii from pixels.

02. Pivot lugs and platform: oblique view



Source: IMG_6559(1).JPG

View. View from the pivot-lug side, slightly above and to the side of the platform. The nearer lug appears larger because of perspective.

Visible features. Both holes, rounded lug crowns, the platform with small tabs and the transition to the wide flange are visible. The paddle recedes into the background.

CAD use. Check how the lugs join the platform. The holes must be coaxial; do not reproduce the apparent size difference between the near and far lugs.

03. Side profile of the part



Source: IMG_4391(1).jpg

View. Near-side view along the hole axis. One lug is seen almost in profile and obscures the other. The part is held by hand; fingers hide part of the paddle end.

Visible features. The paddle extends left, the lug platform is above, and the inclined flange is on the right. The remaining latch tooth projects from the rear at the lower right.

CAD use. Primary view for feature orientation and joint angles. Model the tooth almost straight and essentially parallel to the main paddle plane. This is not a true orthographic view.

04. Rear face: ribs and latch tooth



Source: IMG_6560(1).JPG

View. Oblique rear view from the projecting-tooth side. The tilted part exposes the back of the flange and the underside of the transition.

Visible features. Three reinforcing ribs, an elongated flange recess, two tooth cavities and the damaged area are visible. The lugs are partly visible on the opposite side of the platform.

CAD use. Use to locate ribs, recess and cavities. Do not reproduce cracks or chips. Cavity depths and the missing retaining-face profile cannot be determined from this view.

05. Rear face: second oblique view



Source: IMG_6561(1).JPG

View. The rear face is viewed at a different angle from the preceding photograph. The broad flange face is exposed, with the paddle behind it.

Visible features. The wings, long recess, central tooth and three transition ribs are clear. The central assembly connects to the platform beneath the lugs.

CAD use. Compare with photo 04 to distinguish ribs, raised edges and recesses. Do not interpret highlights or shadows as additional geometric steps without confirmation.

06. Rear view of the flange: overall outline



Source: IMG_4397.jpg

View. Almost face-on view of the rear flange surface, with a slight camera tilt. The part is rotated: the lugs point down in the frame and the tooth lies near the upper convex edge.

Visible features. The full wing span, smooth convex edge, elongated recess, central tooth and ribbed transition are visible. Circular marks and lettering appear on the plate surface.

CAD use. Best available view for the approximate flange outline. Do not apply R27 to the entire edge: a 70 mm chord is impossible for that radius. Do not interpret circular marks as through-holes.

07. View towards the tooth and flange edge



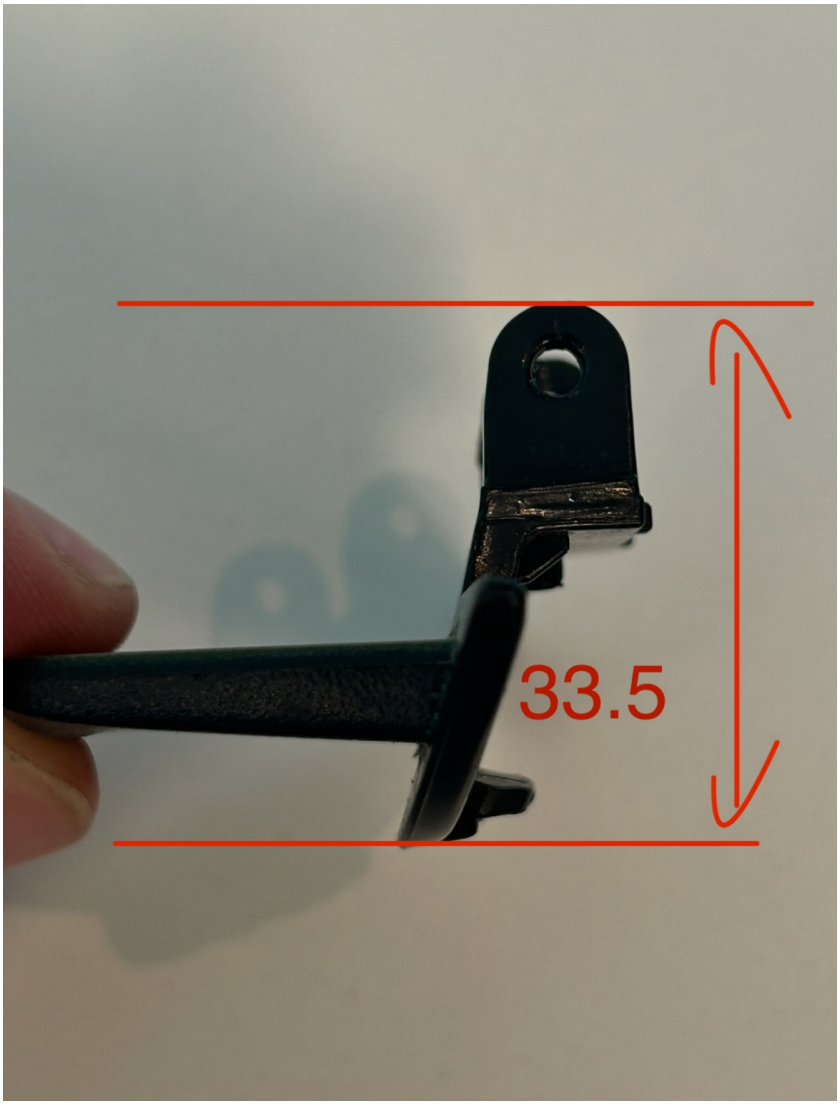
Source: IMG_4393.jpg

View. Low oblique view from the tooth side, almost along the flange face. The broad paddle points down in the frame; the platform tabs are visible behind the tooth.

Visible features. Two tooth cavities and their dividing web are visible centrally. The flange edge extends to either side; below it are the textured paddle face and rounded free end.

CAD use. Use for the tooth cross-section and the flange-to-paddle relationship. A curved edge silhouette in this projection alone does not establish the amount of out-of-plane plate curvature.

08. Overall-height annotation: 33.5 mm



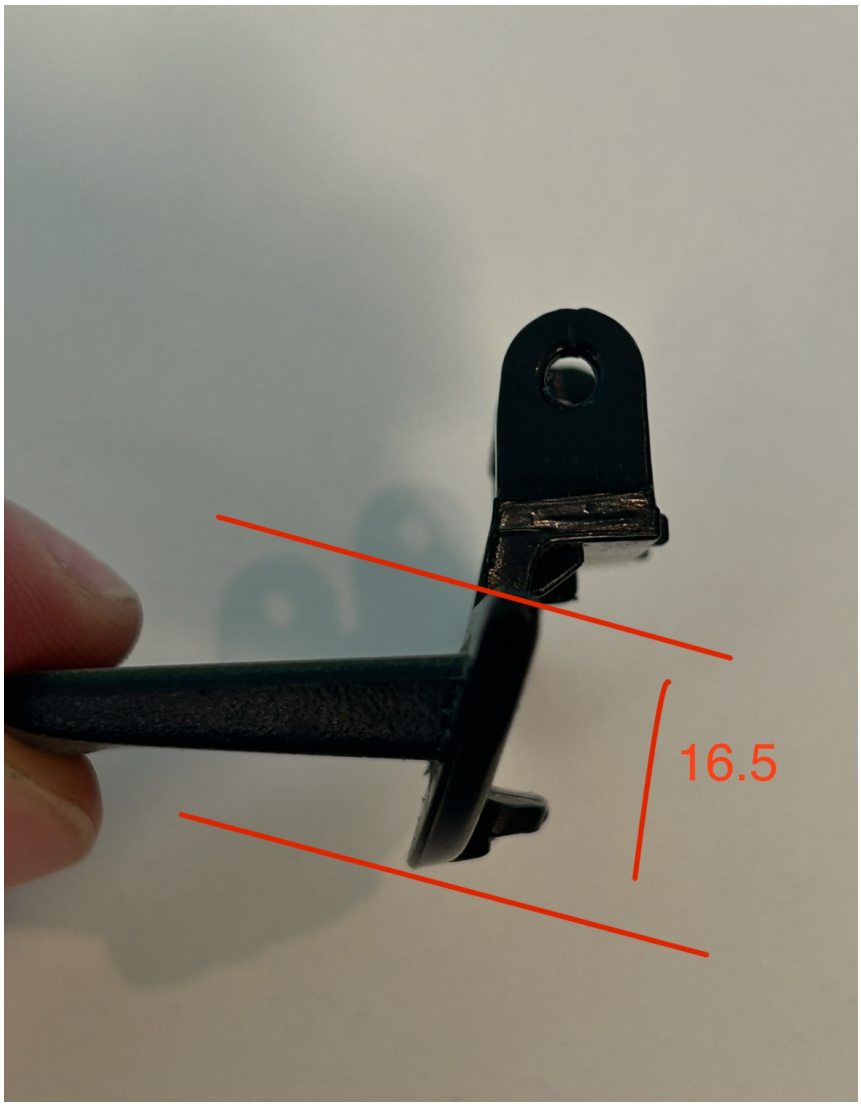
Source: IMG_4394.jpeg

View. The same side view as photo 03, with owner-added markings. Two horizontal red lines indicate the extreme levels of the lug and flange.

Visible features. The 33.5 mm dimension is the envelope from the lug crown to the lowest extreme edge of the support flange. It is not measured to the free end of the paddle.

CAD use. Use as controlling dimension H1 along direction H. The red lines explain the measurement endpoints; the arrow length in pixels is not a separate measurement.

09. Flange-height annotation: 16.5 mm



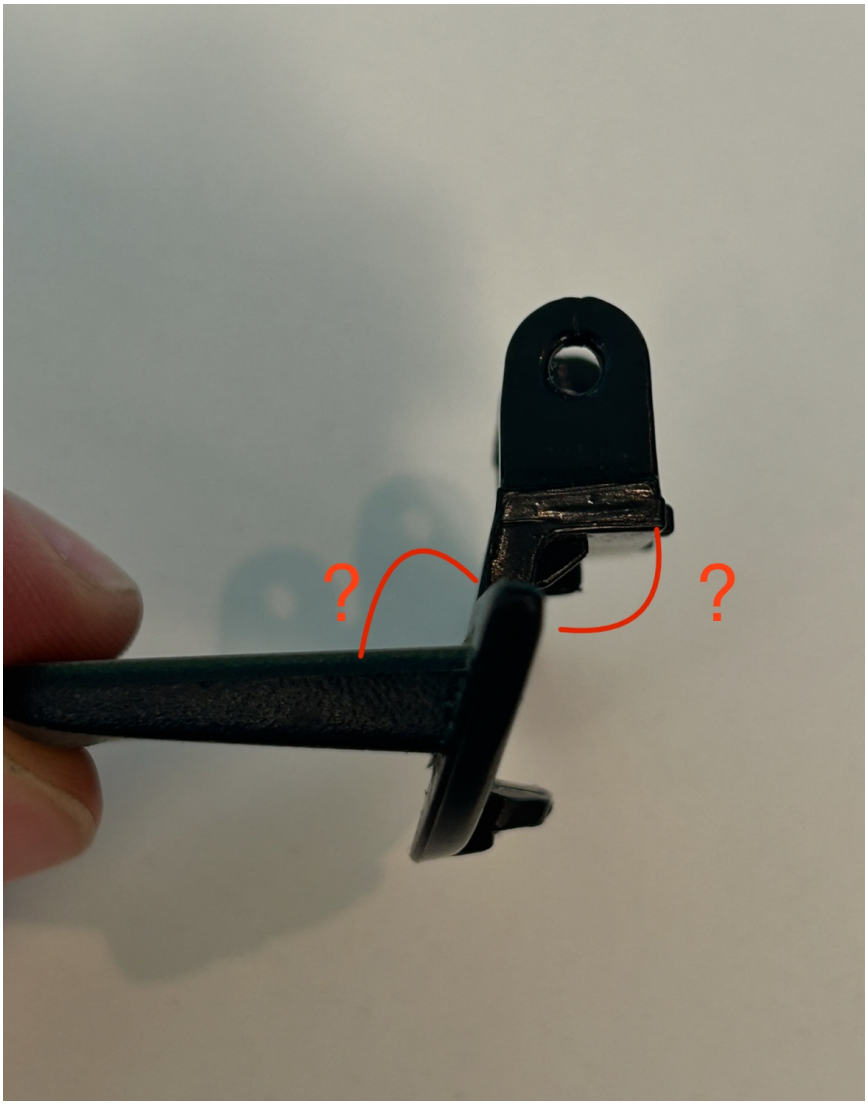
Source: IMG_4395.jpeg

View. Side view with two inclined red reference lines at the extreme edges of the flange itself. The owner drew the lines by hand.

Visible features. The 16.5 mm dimension is the flange height along its inclination. It is neither the 3.2 mm plate thickness nor the 33.5 mm overall envelope.

CAD use. Working CAD interpretation: measure along the main flange direction, approximately normal to the red lines. Do not subtract 16.5 from 33.5 without accounting for their different directions.

10. Features used to estimate the two angles



Source: IMG_4396.jpeg

View. Side view with two red arcs and question marks. The owner did not measure numerical angles; the arcs only identify the joints of interest.

Visible features. The paddle-to-inclined-flange joint is marked on the left, and the platform-to-neck joint on the right. Estimates use straight contour segments, not the shapes of the red arcs.

CAD use. Use approximately 110 degrees for both obtuse angles in the first CAD model. The estimated 105-115 degree range is not a manufacturing tolerance. The rays are defined on the following page.

Comparison with the CAD model

Unambiguous angle definitions

Paddle-to-flange angle, approximately 110 degrees: in the side section, draw one ray from the theoretical intersection towards the paddle free end (left in photo 10), and the other along the flange towards its upper edge and the lugs (up-right). Use the obtuse angle between these rays.

Platform-to-neck angle, approximately 110 degrees: from the platform-to-neck transition, direct one ray along the platform towards the tabs (right in photo 10), and the other along the neck towards the flange (down-left). This is a separate angle; retain an independent parameter.

Tooth direction: align the main longitudinal tooth body essentially parallel to the main paddle plane. Use 0 degrees initially as a working assumption. A local wedge bevel does not imply bending the whole projection.

Feature	Photographs to compare
Paddle	01 and 07: taper, free edge and rounded corners. 03: profile and flange connection.
Pivot lugs	01-03: pair of lugs, crown shape and common hole axis.
Flange	04-07: wings, external outline and long recess. 09: meaning of 16.5 mm.
Transition and ribs	02, 04-06: platform, neck and three reinforcements.
Tooth	03-07: location and surviving surfaces. The damaged face cannot be recovered exactly.
Overall envelope	08: endpoints of 33.5 mm. 10: joint orientations.

Reconciliation rule: owner measurements take priority over photo scaling. The 33.5 and 16.5 mm dimensions have different directions; close the side-sketch chain using outer extreme points. Estimated angles and the reconstructed tooth are not verified fit parameters.

Before manufacture, check hole coaxiality and pin fit, retaining-face position relative to the pivot, free movement and interference with the mating component. Specify material, tolerances and process compensation separately.

Specification A | Datums and dimension status

All dimensions are in millimetres unless stated otherwise. Decimal points replace Russian decimal commas. The dimensions below define a first complete reconstruction, with the remaining uncertainties explicitly retained.

Code	Meaning
M	Owner-supplied or explicitly updated measurement.
S	Value from the earlier specification, retained in the conversation; not independently remeasured.
E	Approximation from photographs or the earlier description.
A	Working assumption adopted to complete the first CAD model.
C	Calculated value, conditional on its stated inputs.

Datum / direction	Definition
B / X	B is the centre symmetry plane; X is the transverse width direction.
C	Common pivot-hole axis.
A	Flat platform surface supporting the pivot lugs.
P	Main paddle reference plane; paddle length is measured along its longitudinal direction.
S	Step where the paddle emerges from the flange.
H / Z	Overall-height direction between the red horizontal reference lines in photo 08. Initially use H = Z.
F	Main flange direction in the side view, from its upper edge towards its lower edge.
V	Local flange-outline coordinate from the line joining the wing tips towards the convex central edge.

For the first model, assume platform A is parallel to paddle plane P, with H normal to them. Keep the platform-to-paddle angle editable. The flange is inclined approximately 20 degrees to H when the selected obtuse joint angle is 110 degrees; match the direction in the side photograph, not its mirror image.

Parameter	Value / interpretation
H1 (M)	33.5: outer envelope along H, lug crown to lowest flange edge; excludes paddle free end.
H2 (M)	16.5: flange size along F between its extreme edges, not plate thickness or vertical projection.
H3 (M)	11.3: lug height normal to A, platform surface to crown; supersedes 10.5 height estimate.
L1 (M/A)	30.0: straight distance along P from step S to paddle free end; replaces 28.5-30.0 estimate.

Specification B | Pivot lugs, platform and neck

Feature	Value / modelling instruction
Lug pair (S)	Two symmetric lugs, each 3.0 thick; clear internal gap 21.8.
Hole diameter (S)	2.5, coaxial through both lugs; no counterbore.
Outer lug span (C)	21.8 + 2 x 3.0 = 27.8. Midplane spacing = 24.8.
Lug height (M)	11.3 from A to crown.
Material above hole (S)	2.9 from hole upper edge to crown.
Hole centre height (C)	11.3 - 2.9 - 1.25 = 7.15 above A. Lower hole edge = 5.90 above A.
Crown-to-hole centre (C)	4.15. This is a distance, not proof of a crown radius.
Lug base length (S)	10.5; initially locate hole halfway along it, at 5.25 (A).
Crown construction (A)	Start with an R4.15 arc centred on the hole, blended tangentially to the 10.5 base. Preserve overall height 11.3.
Lug root / hole edges (A)	Root fillet R0.5, preserving the working 21.8 gap; hole-edge break 0.2 x 45 degrees.
Platform (S)	Width 30.8; thickness 2.5. Initial lug bases flush with platform ends (A).
Lateral remainder (C)	(30.8 - 27.8) / 2 = 1.5 per side, if measured in the same section.
Small platform tabs (S/E)	18 centre spacing; 2 projection; 2.2 width; 2.5 thickness. Exposed edge rounding R0.4 (A). Projection lies in A beyond the tab-side edge (A).
Neck (S/E)	Width transitions from 32.68 to 34; thickness 3.0. Initial straight inclined length 5.1, subordinate to H1 closure.
Platform-to-neck angle (E)	Start at 110 degrees; independent of the other 110-degree angle.
Three reinforcing ribs (E/A)	One on B, outer two near transition side edges. Thickness 2.0; approximate profile 4.5 x 4.5 adapted to the actual angle; root fillets R0.5.

Keep lugs, platform, neck, ribs and flange connected as one valid solid. Neck length 5.1 is a starting reference, not a second rigid constraint that may override the confirmed 33.5 envelope. Show the final dependent neck length on the drawing.

Specification C | Flange and paddle

Feature	Value / modelling instruction
Flange span (S)	70 across the wing tips, after edge rounding.
Flange height (M)	16.5 along F.
Flange thickness (S)	3.2 normal to its local plane.
Central transition width (S/A)	Start at 36 at the wing-tip baseline; blend smoothly into the 34-wide neck. Do not treat this as the short side of an ideal trapezoid.
R27	Do not constrain the whole edge to R27. A 70 chord exceeds the 54 diameter of an R27 circle.
Flange edge rounding (A)	R0.5 on small exposed edges; R1 at wing tips in the outline.
Long rear recess (E/A)	48 x 2.5; depth 1.5. Symmetric in X; closed semicircular ends R1.25. Initial centre at V = 5.0 from the wing-tip baseline.
Recess floor (C)	3.2 - 1.5 = 1.7 only when measured normal to the same plate section. The wider visible border is not automatically part of the 2.5 recess width.
Paddle length (M/A)	30 from S to free edge, measured straight along P.
Paddle widths (S/A)	52 at root; 40 at free end. Initially treat 40 as the tapered blank width before corner rounds.
Paddle outline corners (S/A)	Two free-end corner radii R10 in plane P.
Paddle thickness (S/A)	6 at root, smoothly tapering to 3 at free edge. The 3 value is assumed, not measured.
Paddle end round (A)	R1.5 through thickness.
Internal paddle relief (A)	Do not add a separate deep hollow in the first model; use a smooth thickness profile and rounded end. Hidden-face geometry is not established.

Initially model the flange without separately prescribed out-of-plane curvature. Photo 07 does not distinguish planar outline curvature from spatial bending. If a later comparison requires bending, introduce it as an independent parameter; do not alter outline radius to imitate it.

Omit fine texture, lettering and circular manufacturing marks in the basic model. Do not treat these marks as holes without confirmation.

Specification D | Approximate flange outline

Use local X across the flange width and V from the straight line joining the outer wing tips towards the convex central edge. The origin is the midpoint of that line. The outline is symmetric about X = 0. Local top/bottom is independent of photograph rotation.

X	V	Status
0	16.5	Measured maximum; local outline placement assumed.
+/-9	16.2	Estimated / assumed.
+/-18	13.8	Estimated / assumed.
+/-26	8.5	Estimated / assumed.
+/-35	0	70 span from earlier specification; local baseline assumed.

Pass a smooth symmetric spline through these initial points without waves or inflections. At the centre, the tangent is parallel to X. Compare visually with photo 06; intermediate points may be adjusted while retaining the specified envelope. These points are a starting reconstruction, not a coordinate survey.

Initially use V = 0 as the opposite wing edge, blending into the central neck. Use 36 as the starting width of this central connection, and transition smoothly to the 34-wide neck. Preserve the final 70 span after rounding.

Controlling the height chain

H1 = 33.5 is the controlling envelope. Preserve lug height 11.3, flange size 16.5 and their thicknesses. Measure H1 between support planes tangent to the extreme external points, not by adding centreline lengths. Adjust transition location/effective neck length to close H1. Record the resulting neck dimension.

With a 20-degree inclination, the thin-line projection of the flange height on H is 16.5 cos(20 degrees) = 15.51. The remainder 33.5 - 11.3 - 15.51 = 6.69 is a combined base/transition interval, not a standalone neck length.

The simplified chain 11.3 + 2.5 + 5.1 cos(20 degrees) + 16.5 cos(20 degrees) is about 34.10, not 33.5. It may double-count overlapping transitions. Resolve the approximately 0.6 difference through actual CAD junction geometry and the dependent transition length; do not reduce the confirmed lug height.

Specification E | Latch-tooth reconstruction

The tooth is symmetric about B, with overall width 17.5 (S). Its main longitudinal direction is straight and parallel to P. It projects from the rear of the flange. Avoid a pronounced hook; a local wedge bevel does not bend the whole tooth.

Parameter	Starting model
Root location (A)	Near the lower/convex outer flange edge, as in the side view. Initially leave 0.5 along F from that edge to the closest tooth-root boundary.
Projection (S)	10 along the tooth direction from the agreed rear root surface.
Maximum thickness (S/A)	5.2 normal to P, initially at the root. The earlier description did not unambiguously distinguish root thickness from a retaining step.
Free-end thickness (S)	2.0.
Envelope (A)	Keep one longitudinal face straight and parallel to P; taper the opposite face. This is an initial wedge reconstruction, not an established original profile.
Root fillet (A)	R1.0 outside the contact region.
End-edge break (A)	R0.3.
Internal cavities (A)	Use a solid tooth for the first reconstruction. The sample has two cavities, but their dimensions and purpose are not established. A solid tooth is not equivalent in stiffness to the original.

Do not assign an arbitrary retaining-step depth and present it as measured. Keep the working retaining face parametrically editable for positioning against the mating component. Until that check, its shape and position remain unverified.

Approximate position check from the earlier photo assessment

In IMG_4394.jpeg, the earlier assessment placed the visible hole centre near pixel (775,510) and the tooth root near (650,1100). Using a rough scale of 756 / 33.5 = 22.6 pixels/mm, the root is about 26 below C along H and 5-6 towards the paddle in the image horizontal direction. These are rough orientation checks with uncertainty of several millimetres, not hard sketch constraints or retaining-face coordinates.

Do not copy fracture surfaces. If matching the original cavity structure is required, obtain cavity dimensions rather than deriving them from shadows. Check movement, clearance, engagement and required flexibility before accepting a solid-tooth version.

Specification F | Angle estimates and consistency

Parameter	First-model value / status
Alpha: paddle to flange	110 degrees (E); visual estimate range 105-115 degrees. Use the obtuse rays defined on page 12.
Beta: platform to neck	110 degrees (E), separately editable; visual estimate range 105-115 degrees.
Gamma: tooth to P	0 degrees (A), following the owner requirement of near parallelism.
Delta: A to P	0 degrees (A); approximate parallelism is not a measured equality.

The 105-115 degree range describes reconstruction uncertainty, not a manufacturing tolerance. Approximately 70 degrees is the supplementary acute angle between the same lines using different rays.

Traceability of the previous angle assessment

The earlier assessment used IMG_4396.jpeg at 1170 x 1490 pixels, with origin at the top left: paddle segment (200,893)-(610,878); flange (590,1130)-(697,824); platform (680,711)-(891,695); neck (638,822)-(671,705). Angles between the corresponding ray vectors were approximately 111 and 110 degrees. These are manually selected contour segments and projected angles, not exact spatial measurements.

These calculations are retained from the earlier specification discussion, not presented as new metrology. Rounded junctions, perspective and selection of straight edges limit accuracy.

Consistency check	Result
Lug outside width	21.8 + 2 x 3.0 = 27.8.
Lug midplane spacing	21.8 + 3.0 = 24.8.
Crown to hole centre	2.9 + 2.5 / 2 = 4.15 (distance).
Hole centre above A	11.3 - 4.15 = 7.15.
Hole bottom above A	7.15 - 1.25 = 5.90.
R27 / 70 chord	Incompatible for a single circular arc: 70 > 54.

The previous side-photo cross-check estimated 756 pixels for the 33.5 envelope and approximately 255-260 pixels for lug height, corresponding to roughly 11.3-11.5. This is only a coarse agreement check; the paper grid and raised part do not share the same image scale.

Specification G | Modelling and handover

Recommended construction sequence

1. Create the parametric side sketch and symmetry plane B. Main inputs: H1 = 33.5, H2 = 16.5, H3 = 11.3, L1 = 30. Initial angles: alpha = beta = 110 degrees; gamma = delta = 0 degrees.

2. Build the platform, tabs and lugs. Keep hole diameter 2.5, lug thickness 3 and internal clearance 21.8 as separate editable parameters.

3. Build the symmetric flange outline, thickness and inclination. Position it to produce an external H envelope of exactly 33.5. Connect it to the platform with a neck whose effective length is subordinate to that envelope.

4. Build the paddle from S and the rear tooth. Confirm that the tooth is almost straight and parallel to P, rather than curled towards the flange.

5. Add ribs, recess and fillets. Produce one valid solid without gaps, self-intersections or disconnected features. Omit damage.

6. Compare side, front and rear model views with the photographs. Do not change confirmed dimensions to remove a mismatch caused by perspective.

Files and drawing to deliver

Supply the native parametric CAD file, STEP and a drawing containing a side section, flange view, lug profile and assumption table. For printing, add STL or 3MF in millimetres. Show final dependent neck length and tooth location relative to axis C.

Acceptance criteria for the first geometry model

Preserve the four owner-updated dimensions 33.5 / 16.5 / 11.3 / 30 using the defined datums. Make both holes coaxial; distinguish internal gap from centre spacing. Use a smooth symmetric 70-wide flange outline without applying R27 across the full span. Build the correct obtuse-angle orientation. Keep the tooth nearly straight. Mark estimates and assumptions in the parameters.

Checks before final manufacture

Check pin fit, retaining-face position relative to the pivot, lever travel, flange/tab interference and required latch force against the mating component or a prototype. Material, manufacturing tolerances and printing compensation remain to be specified. If only one further check is possible, prioritise trial engagement of the reconstructed tooth with its mating catch.

This package supports a first complete CAD reconstruction. It does not establish exact functional interchangeability: the missing tooth profile and uncertain dimensions require physical validation.